Hierarchical Logistic Regression — Summary

As an additional in-depth analysis, a hierarchical logistic regression model was conducted. Variables were entered in stages based on theoretical blocks: substance-use profile, socioeconomic variables, demographic variables, mental health, and finally treatment and legal-history variables. The goal was to assess whether each group of variables contributed additional explanatory power beyond those already included in the model.

The most comprehensive model, which included lifetime users of heroin, cocaine, and methamphetamine, demonstrated the best and most stable performance. This model achieved a ROC-AUC of 0.750 and an F1 score of 0.287, the highest among the three hierarchical models. However, precision remained relatively low across all models, likely due to class imbalance in the outcome variable.

Model	Preferred model	ROC-AUC	Recall / Sensitivity	Precision	F1	Balanced Accuracy
Heroin	Model 1 / SES	0.697	0.671	0.108	0.186	0.660
Heroin + Cocaine	Model 3	0.699	0.731	0.092	0.163	0.647
Heroin + Cocaine + Meth	Model 3	0.750	0.630	0.186	0.287	0.683

Overall, the hierarchical model supports the main conclusion of the project: active past-month use among lifetime users is associated with a combination of substance-use profile, socioeconomic vulnerability, and mental health status. The strongest and most stable model was the combined heroin/cocaine/methamphetamine model, suggesting that broader substance-use profiles provide useful information for identifying active use, while mental health and socioeconomic factors remain central explanatory components.